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Exploring the Bell State

Introduction

Quantum mechanics, the foundation of modern physics, introduces concepts that challenge classical intuitions. One of these concepts is entanglement, a phenomenon where particles become interconnected in such a way that the state of one particle is dependent on the state of another, regardless of the distance separating them. Within this framework, Bell states are paramount as they represent the simplest and most profound examples of entanglement. This research paper explores the origin, historical context, properties, and applications of Bell states, highlighting their significance in quantum mechanics.

Origin and Historical Context

The journey to understanding Bell states begins with the Einstein-Podolsky-Rosen (EPR) paradox, proposed in 1935. Albert Einstein, Boris Podolsky, and Nathan Rosen questioned the completeness of quantum mechanics, suggesting that there must be hidden variables governing the behaviour of particles. They posited that if quantum mechanics were complete, it would allow "spooky action at a distance," which Einstein found unsettling.

In 1964, physicist John Bell formulated Bell's theorem, which provided a way to test the EPR paradox. Bell derived inequalities that local hidden variable theories must satisfy. Quantum mechanics, however, predicted violations of these inequalities under certain conditions. Bell's theorem thus laid the groundwork for experimental tests to distinguish between classical and quantum predictions.

One of the most notable experiments verifying Bell's theorem was conducted by Alain Aspect in the 1980s. Aspect's experiments demonstrated that entangled particles indeed violate Bell's inequalities, supporting the quantum mechanical description and ruling out local hidden variable theories. These results underscored the reality of quantum entanglement and the existence of Bell states.

Properties and Applications of Bell States

Bell states are four specific maximally entangled quantum states of two qubits, denoted as $|\Phi+\rangle$, $|\Phi-\rangle$, $|\Psi+\rangle$, and $|\Psi-\rangle$. These states are described mathematically as:

$$|\Phi+\rangle = (1/\sqrt{2})(|0\rangle A \otimes |0\rangle B + |1\rangle A \otimes |1\rangle B)$$

$$|\Phi-\rangle = (1/\sqrt{2})(|0\rangle A \otimes |0\rangle B - |1\rangle A \otimes |1\rangle B)$$

$|\Psi+\rangle = (1/\sqrt{2})(|0\rangle A \otimes |1\rangle B + |1\rangle A \otimes |0\rangle B)$

$|\Psi-\rangle = (1/\sqrt{2})(|0\rangle A \otimes |1\rangle B - |1\rangle A \otimes |0\rangle B)$

They are known as the four *maximally entangled two-qubit Bell states* and form a maximally entangled basis, known as the Bell basis, of the four-dimensional Hilbert space for two qubits The result of measuring a single qubit in a Bell state is uncertain. However, if you measure the first qubit in the z-basis, the second qubit's measurement will always match the first qubit's value (for $|\Phi+\rangle$ and $|\Phi-\rangle$ Bell states) or be the opposite (for $|\Psi+\rangle$ and $|\Psi-\rangle$ Bell states). This shows that their outcomes are correlated. John Bell demonstrated that these correlations are stronger than what classical systems can achieve, suggesting that quantum mechanics enables information processing beyond classical limits. Bell states form an orthonormal basis, meaning they can be uniquely identified through measurement. As entangled states, Bell states provide information about the entire system but not about the individual qubits. For instance, while a Bell state is a pure state, the reduced density operator for one qubit is a mixed state, indicating incomplete information about that single qubit. Bell states can be either symmetric or antisymmetric and are maximally entangled, meaning their reduced density operators are maximally mixed. The multipartite generalization of Bell states, with similar maximal entanglement properties, is known as the absolutely maximally entangled (AME) state.

Applications

1. **Quantum Cryptography :** Bell states enable secure communication through quantum key distribution (QKD). QKD protocols like BB84 use entanglement to detect eavesdropping, ensuring the confidentiality of the shared key.

2. **Quantum Teleportation :** Bell states facilitate the transfer of quantum information between distant parties. By entangling a pair of qubits, a quantum state can be "teleported" from one location to another, a process crucial for quantum communication and computing.

3. **Superdense Coding :** This technique allows the transmission of two classical bits by sending only one qubit. Utilizing entangled states, superdense coding enhances the efficiency of data transmission in quantum communication networks.

Implementation in Code

In [1]:
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%pip install qiskit[visualization]==1.1.0
```

```
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3.post1)
Requirement already satisfied: tzdata>=2022.1 in c:\users\arka singha\anaconda3\lib
```

In [2]:
```
import qiskit
```

In [3]:
```
qiskit.__version__
```

Out[3]: '1.1.0'

In [5]:
```
%pip install qiskit_aer
%pip install qiskit_ibm_runtime
%pip install matplotlib
%pip install pylatexenc
%pip install qiskit-transpiler-service
```

In [12]:
```
pip install qiskit-aer
```

In [23]:
```
from qiskit import QuantumCircuit, ClassicalRegister, QuantumRegister, transpile
from qiskit_aer import Aer
from qiskit.visualization import plot_histogram

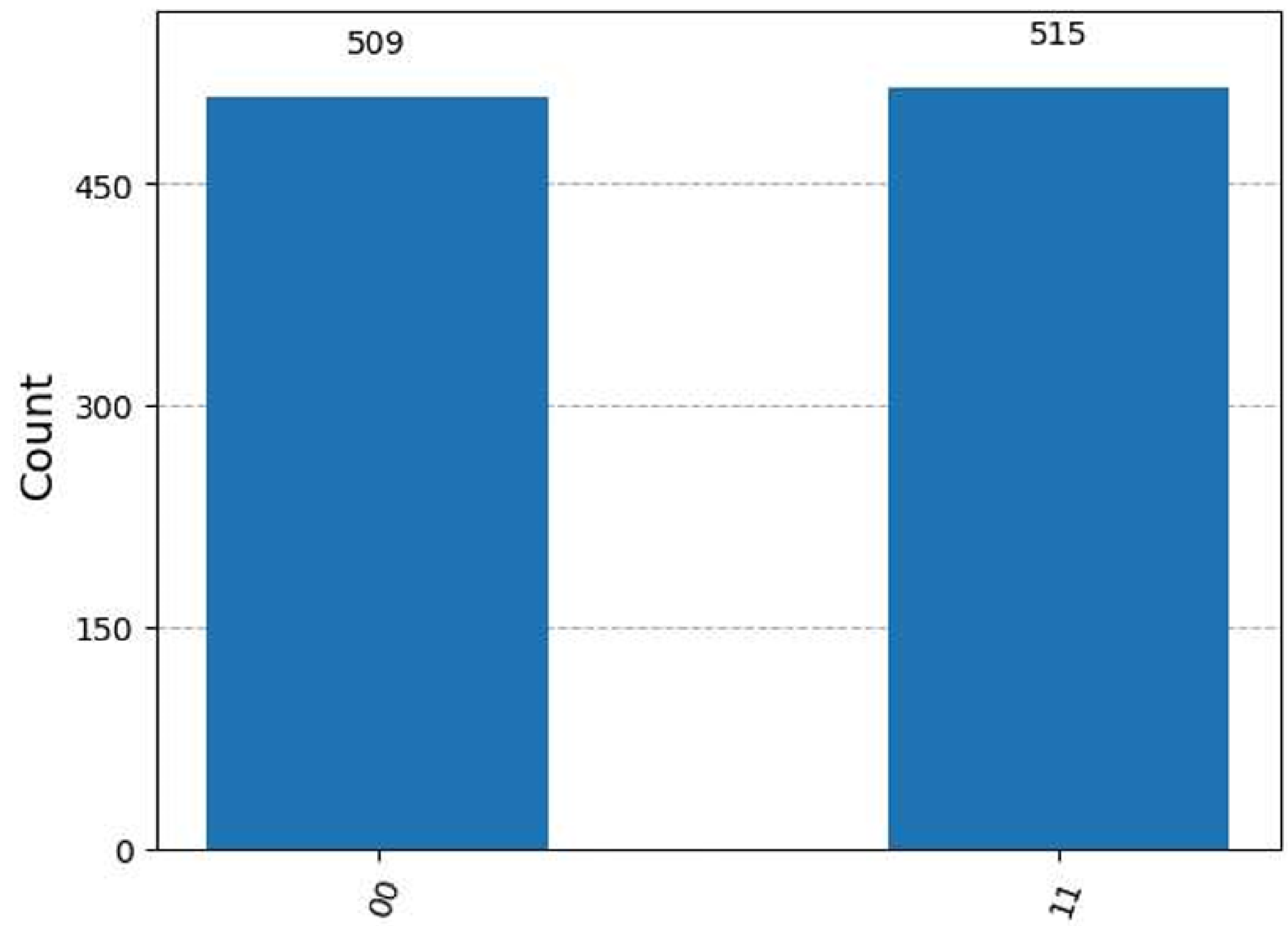
qr = QuantumRegister(2)
cr = ClassicalRegister(2)

circuit = QuantumCircuit(qr, cr)
circuit.h(qr[0])
circuit.cx(qr[0], qr[1])
circuit.measure(qr, cr)

simulator = Aer.get_backend('qasm_simulator')
transpiled_circuit = transpile(circuit, backend=simulator)
job = simulator.run(transpiled_circuit, shots=1024)

result = job.result()
counts = result.get_counts(transpiled_circuit)
plot_histogram(counts)
```

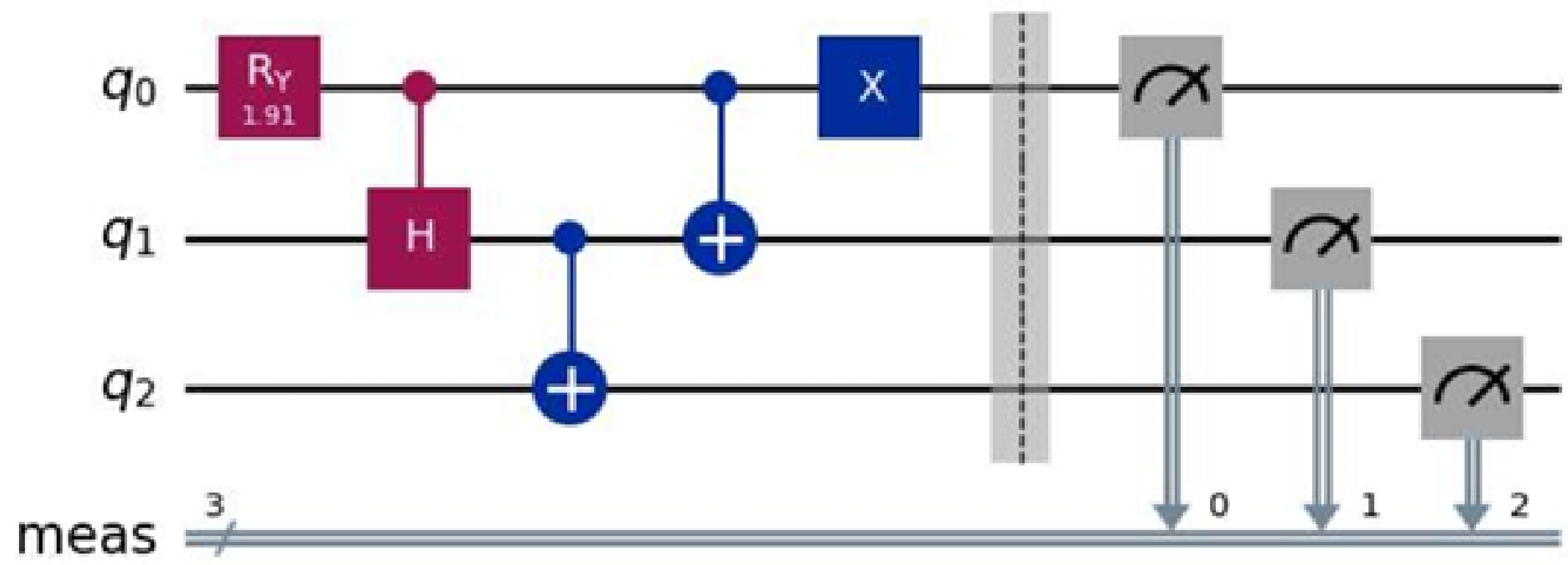
Out[23]:



In [36]:

```
#Singlet Bell state circuit
from qiskit import QuantumCircuit
qc = QuantumCircuit(3)
qc.ry(1.91063324, 0)
qc.ch(0,1)
qc.cx(1,2)
qc.cx(0,1)
qc.x(0)
qc.measure_all()
qc.draw('mpl')
```

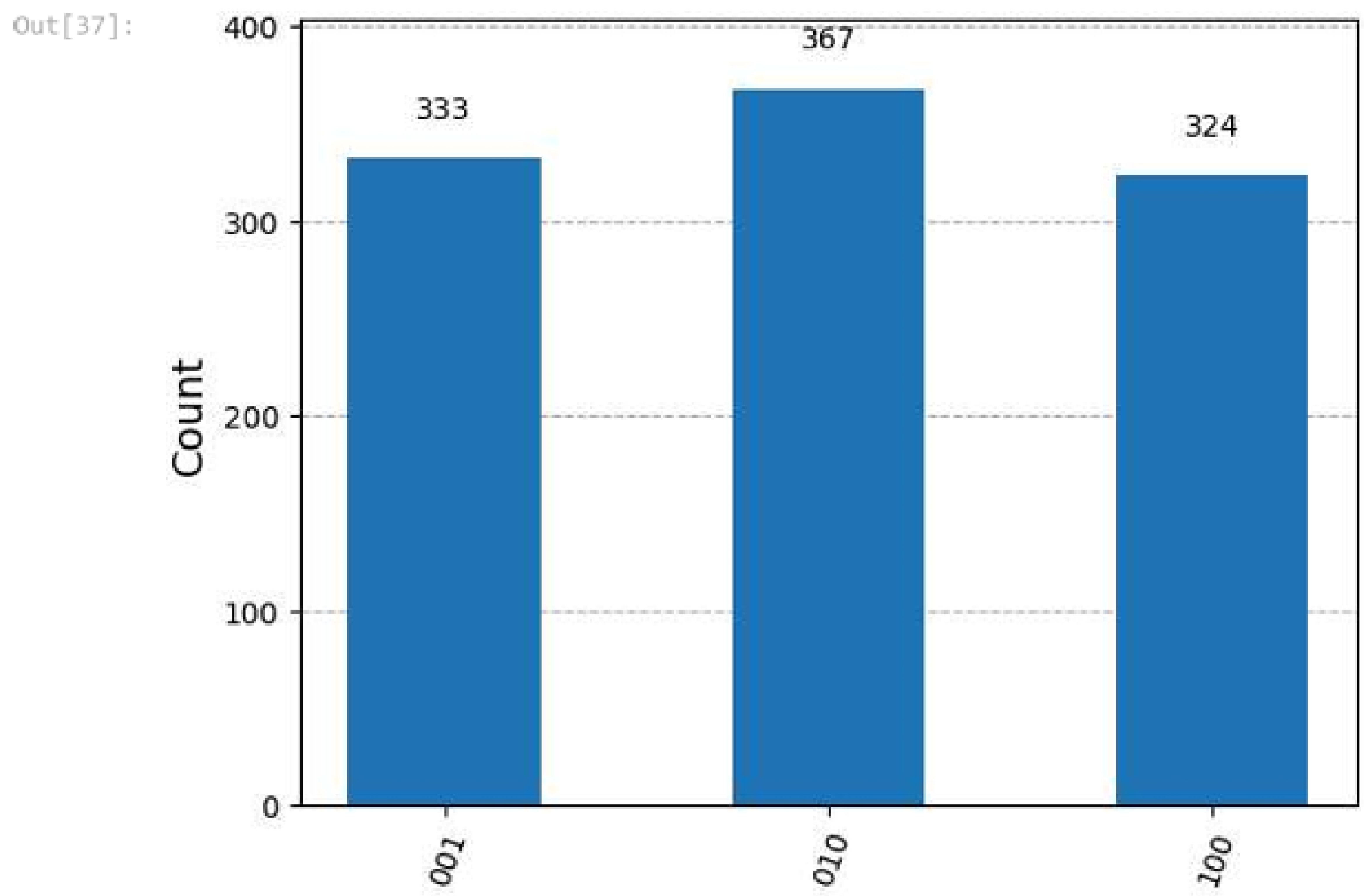
Out[36]:



In [37]:

```
#Weil's Inequality
sampler = StatevectorSampler()
pub = (qc)
job_sampler = sampler.run([pub], shots=1024)
result_sampler = job_sampler.result()
counts_sampler = result_sampler[0].data.meas.get_counts()
print(counts_sampler)
plot_histogram(counts_sampler)
```

```
{'010': 367, '100': 324, '001': 333}
```



Conclusion

Bell states are a cornerstone of quantum mechanics, exemplifying the non-classical nature of entanglement. From their historical roots in the EPR paradox and Bell's theorem to their verification in experiments by Aspect and others, Bell states have revolutionized our understanding of quantum phenomena. Their unique properties enable groundbreaking applications in quantum cryptography, teleportation, and superdense coding. As research in quantum information science advances, Bell states will continue to play a pivotal role in harnessing the full potential of quantum technologies.

References

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3. Einstein, A., Podolsky, B., & Rosen, N. (1935). Can Quantum-Mechanical Description of Physical Reality Be Considered Complete? **Physical Review, 47**(10), 777-780.

4. Nielsen, M. A., & Chuang, I. L. (2000). **Quantum Computation and Quantum Information**. Cambridge University Press.